Individual Assignment Report

Theme: Interactive Data Visualizations in React and D3.js

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1. Characterization of the Problem, the Data and the User Tasks

The problem is to help users decide **which U.S. state or city is better to settle down in**, based on safety and economic factors. The dataset used is the "Communities and Crime" dataset from UCI, which contains socio-economic and crime-related attributes for thousands of communities across the United States.

Key data attributes used:

- medIncome: Median household income (economic well-being)
- ViolentCrimesPerPop: Violent crime rate per population (safety level)
- state and communityname: Hierarchical structure
- population: Used for sizing in the treemap

User Goals:

- Find communities with **high income** and **low violent crime rate**
- Understand the distribution and patterns across states
- Compare multiple communities quickly
- Make an informed decision about where to live

Required Operations:

- Explore multivariate relationships (income vs crime)
- Select multiple communities for comparison (brush selection)
- Navigate hierarchical structure (state → community)
- Get detailed information on demand (tooltip + selection panel)

2. Design Rationale

I implemented two coordinated visualizations to support the user's decision-making process:

a) Scatterplot with 2D Brush (left)

- X-axis: medIncome (economic factor)
- Y-axis: ViolentCrimesPerPop (safety factor)
- Each point represents one community.

- 2D Brush allows users to select a rectangular region of interest. Selected points are highlighted in red.

b) Treemap (right)

- Two-level hierarchy: State → Community
- Rectangle size encodes population (larger = more populous)
- Color encodes violent crime rate (dark red = high crime, light green = low crime)
- Clicking a rectangle selects that community.

Interactions & Coordination Brushing on the scatterplot highlights corresponding communities in the treemap with thick black borders and full opacity. Clicking a treemap rectangle highlights the matching points in the scatterplot. A top information panel provides real-time statistics including number of selected communities, involved states, average crime rate, and average income.

Why Treemap? I tested three hierarchical layouts proposed by D3.js: Treemap, Icicle, and Sunburst.

- **Treemap**：Excellent space utilization, intuitive size comparison (population), and effective color encoding. It clearly shows both hierarchy and quantitative values. *Cons*：Small communities become very tiny rectangles, making labels difficult.
- **Icicle**：Better for understanding hierarchy levels clearly. *Cons*：Wastes horizontal space and is less visually engaging for size comparison.
- **Sunburst**：Visually attractive for radial exploration. *Cons*：Harder to compare sizes accurately and less suitable when there are many communities.

Preferred Layout: Treemap. It best balances space efficiency, population comparison, and crime rate visualization, directly supporting the user's goal of finding safe and prosperous communities. The coordinated brushing and clicking create a linked-view system that helps users quickly discover patterns, such as clusters of high-income, low-crime communities in certain states.

3. Declaration of Generative AI Use

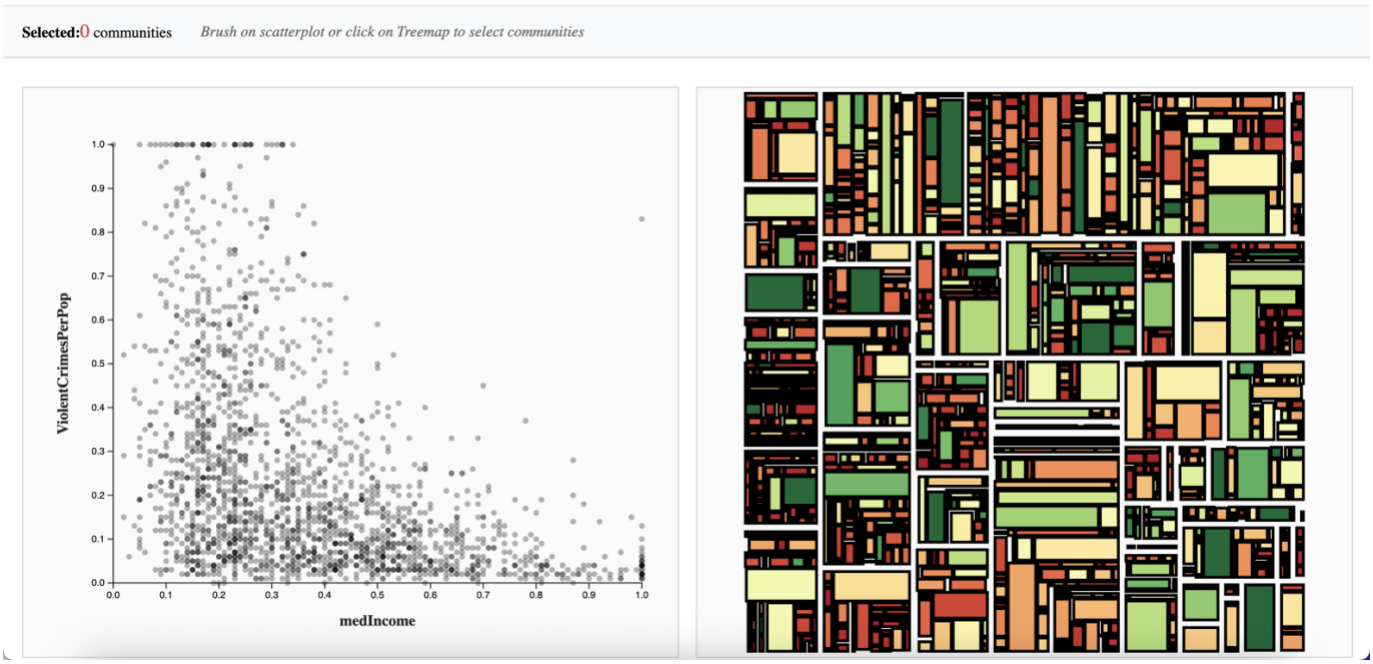
I used **Grok 4.20** extensively during development.

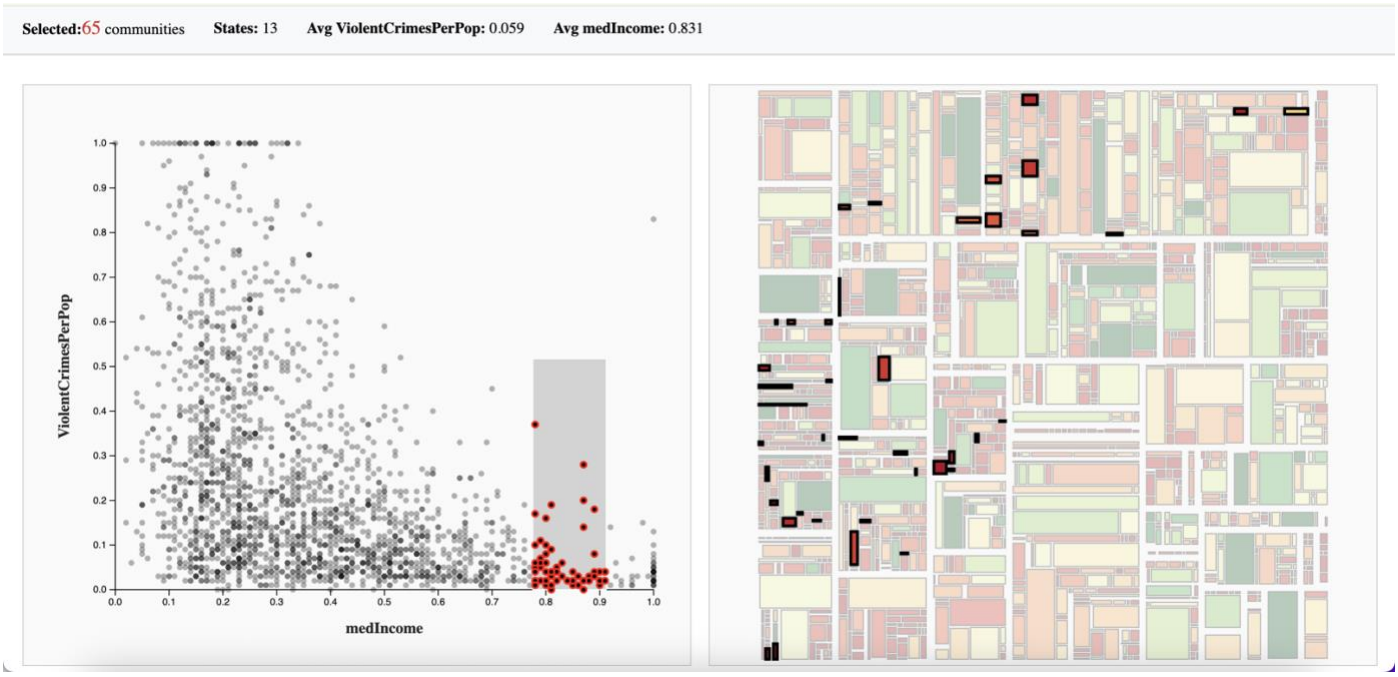
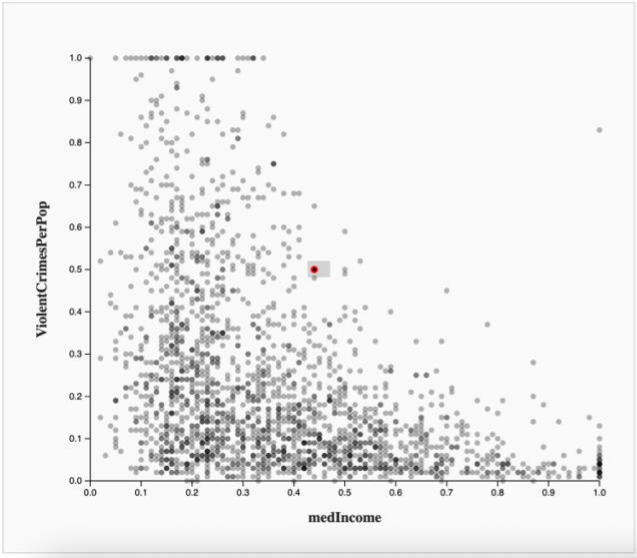
What I used Grok for:

- Generating initial structure for ScatterplotContainer.jsx and HierarchyContainer.jsx
- Debugging Redux state management and synchronization issues
- Writing and refining the Treemap layout in HierarchyD3.js
- Creating the information panel and tooltip
- Helping with layout adjustments and bug fixing

What I did myself:

- Decided the visual design (scatterplot + treemap)
- Chose attributes (medIncome vs ViolentCrimesPerPop)
- Adjusted margins, colors, and highlighing styles
- Tested and compared different hierarchical layouts
- Wrote the report content and rationale





Selected: 1 communities **States:** 1 **Avg ViolentCrimesPerPop:** 0.500 **Avg medIncome:** 0.440